UE26MA141A Engineering Mathematics – I · Unit 2 · Higher-Order Linear Differential Equations

Important points

Definitions, statements, key points, "why" reasoning and exam traps, topic by topic.

Class 002 · Solutions of homogeneous DE: complementary functions · compiled 15 October 2026

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S002·7	Unit 2 deck 002, slide 7 (rendered .pptx)
T002 00:09:10	Unit 2 AV-summary transcript 002 ("Rules for finding the Complementary Function, Problems"), segment start
CW U2 P8	Unit 2 class-work problem 8
ESA Jan-24 s4 2c	Past paper: UE23MA141A Jan 2024 set 4, question 2.c

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1. 1 · The auxiliary equation and the complementary function
2. 2 · Second-order equations: three cases
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5. Past-paper questions

1 The auxiliary equation and the complementary function

Definitions

Homogeneous linear DE with constant coefficients. $(D^n + k_1D^{n-1} + k_2D^{n-2} + \cdots + k_n)y = 0$, written $f(D)y = 0$, where $D = \frac{d}{dx}$ and $k_1, \ldots, k_n$ are constants. S002·10 · T002 00:12:58

Auxiliary polynomial. $f(m)$: replace D by m in the linear differential operator $f(D)$. S002·6 · T002 00:04:00

Auxiliary equation (characteristic equation). $f(m) = 0$. For $\frac{d^2y}{dx^2} + k_1\frac{dy}{dx} + k_2y = 0$ it is $m^2 + k_1m + k_2 = 0$. S002·5 · S002·6 · T002 00:03:01 · T002 00:04:23

Complementary function (CF). The general solution of the homogeneous linear DE $f(D)y = 0$. It contains as many arbitrary constants as the order of the equation. T002 00:00:41 · T002 00:06:31

Statements in the class material

> *"Substituting into the above equation, we see that* $y = e^{mx}$ *is a solution if* $(m^2 + k_1 m + k_2)e^{mx} = 0$ *.* $m^2 + k_1 m + k_2 = 0$ *is the AUXILIARY EQUATION of* $(D^2 + k_1 D + k_2)y = 0$*."* S002·5

> *"The solution of the differential equation depends on the nature of roots of the auxiliary equation."* S002·6

Key points

1. Every non-homogeneous ESA question in Unit 2 (24 of 24 occurrences) needs the CF first; a slip in the AE loses the whole question. ESA ★★★ ESA Dec-23 3a · ESA Jan-24 s2 2d · ESA Jan-24 s4 2c
2. Writing the AE: replace D by m in $f(D)$, or in the DE replace y by 1, $\frac{dy}{dx}$ by m, $\frac{d^2y}{dx^2}$ by m^2, $\frac{d^3y}{dx^3}$ by m^3, and equate to 0. ESA ★★★ T002 00:03:36 · T002 00:07:40
3. $e^{mx} \neq 0$ for every real x, so the bracket $f(m)$ must be 0. That is why only the roots of the AE matter. ESA ★★ S002·5 · T002 00:02:47
4. An equation of order n has n arbitrary constants in its general solution; count them as a check. ESA ★★★ T002 00:06:31 · T002 00:15:54

Why?

Why does the CF matter when the exam only asks non-homogeneous equations?
The general solution of $f(D)y = X$ is $y = \mathbf{CF} + \mathbf{PI}$. The CF is the general solution of $f(D)y = 0$, so every question in classes 003–015 starts here.

T002 00:00:41 · ESA Dec-23 3a

Exam traps

Exam trap. The AE for $\frac{d^3y}{dx^3}$ needs m^3. The lecture says "replace it by m square" at T002 00:07:49; that is a slip of the tongue. ADDED FOR CLARITY T002 00:07:49

Exam trap. A missing y term means a zero root: $m^3 + 4m = m(m^2 + 4)$. Do not divide by m and lose the constant c_1. ADDED FOR CLARITY S002·14

Exam trap. With a leading coefficient, keep it in the factorisation: $4m^2 - 8m + 3 = (2m - 1)(2m - 3)$ gives $m = \frac{1}{2}, \frac{3}{2}$ (CW P4). ADDED FOR CLARITY CW U2 P4

2 Second-order equations: three cases

Statements in the class material

> *"CASE I. If the roots m_1 and m_2 are real and distinct, then the general solution of the differential equation $\frac{d^2y}{dx^2} + k_1\frac{dy}{dx} + k_2y = 0$ is $y = c_1e^{m_1x} + c_2e^{m_2x}$."* S002·7

> *"CASE II. If the roots m_1 and m_2 are real and equal, then the general solution of the differential equation $\frac{d^2y}{dx^2} + k_1\frac{dy}{dx} + k_2y = 0$ is $y = (c_1 + c_2x)e^{mx}$."* S002·8

> *"CASE III. If the roots m_1 and m_2 are complex, let the roots be $m_1 = \alpha + i\beta$ and $m_2 = \alpha - i\beta$; then the general solution of the differential equation is $y = e^{\alpha x}(c_1\cos\beta x + c_2\sin\beta x)$."* S002·9

Key points

1. Classify the roots before writing anything: real and distinct, real and equal, or complex. The quadratic formula settles it. ESA ★★★ S002·6 · T002 00:04:41 · T002 00:11:56
2. In case III, α is the real part and β the imaginary part of the root: $2 \pm 3i$ gives $e^{2x}(c_1\cos 3x + c_2\sin 3x)$. ESA ★★★ S002·9 · T002 00:12:13
3. Complex roots of an AE with real coefficients occur in conjugate pairs $\alpha \pm i\beta$, so they always come two at a time. ESA ★★ T002 00:10:38

Why?

Why $(c_1 + c_2x)e^{mx}$ and not $c_1e^{mx} + c_2e^{mx}$ for equal roots?
$c_1e^{mx} + c_2e^{mx} = (c_1 + c_2)e^{mx}$ has only one independent constant, but a second-order equation needs two. xe^{mx} is a second, independent solution: for $(D - m)^2y = 0$, $(D - m)(xe^{mx}) = e^{mx}$ and $(D - m)e^{mx} = 0$.

ADDED FOR CLARITY S002·8 · T002 00:06:31

Where do $\cos\beta x$ and $\sin\beta x$ come from?
$e^{(\alpha\pm i\beta)x} = e^{\alpha x}(\cos\beta x \pm i\sin\beta x)$ (Euler's formula). Suitable combinations of the two complex solutions give the real solutions $e^{\alpha x}\cos\beta x$ and $e^{\alpha x}\sin\beta x$.

ADDED FOR CLARITY S002·9

Exam traps

Exam trap. Do not write $c_1e^{mx} + c_2e^{mx}$ for a double root: it is $(c_1 + c_2x)e^{mx}$. ADDED FOR CLARITY S002·8

Exam trap. For $2 \pm 3i$, $\beta = 3$, not 2. The lecture first says "beta is 2", then corrects it to 3 (T002 00:12:13); slide 002·9 has $\cos 3x$, $\sin 3x$. S002·9 · T002 00:12:13

Exam trap. Do not put i in the CF: $e^{2x}(c_1 \cos 3x + c_2 \sin 3x)$, never $e^{(2+3i)x}$ terms.

ADDED FOR CLARITY S002·9

Exam trap. Roots $\pm 2i$ have $\alpha = 0$: the CF is $c_1 \cos 2x + c_2 \sin 2x$, with no e^{2x}.

ADDED FOR CLARITY S002·14

3 Equations of order n: combining the cases

Definitions

Multiplicity of a root. The number of times a root is repeated in the AE: in $(m-1)^2(m+1)^2=0$ both 1 and -1 have multiplicity 2. (added for clarity) S002·12 · T002 00:14:47

Statements in the class material

> *"Case (i). If the roots $m_1, m_2, \ldots, m_n$ are real and distinct, then the general solution is $y = c_1e^{m_1x} + c_2e^{m_2x} + \cdots + c_ne^{m_nx}$."* S002·11

> *"Case (ii). If the roots $m_1, m_2, \ldots, m_n$ are real and equal (say equal to m), then the general solution is $y = (c_1 + c_2x + \cdots + c_nx^{n-1})e^{mx}$."* S002·12

> *"Case (iii). If the roots m_1, m_2, m_3, m_4 are real and equal (say equal to m) and the remaining $(n-4)$ roots are real and distinct, then the general solution is $y = (c_1 + c_2x + c_3x^2 + c_4x^3)e^{mx} + c_5e^{m_5x} + c_6e^{m_6x} + \cdots + c_ne^{m_nx}$."* S002·12

> *"Case (iv). If there are two pairs of imaginary roots $\alpha \pm i\beta$, $\gamma \pm i\delta$, then the general solution is $y = e^{\alpha x}(c_1\cos\beta x + c_2\sin\beta x) + e^{\gamma x}(c_3\cos\delta x + c_4\sin\delta x) + c_5e^{m_5x} + \cdots + c_ne^{m_nx}$."* S002·13

> *"Case (v). If a pair of complex roots say $\alpha \pm i\beta$ occurring twice, then the GS is $y = e^{\alpha x}\big((c_1 + c_2x)\cos\beta x + (c_3 + c_4x)\sin\beta x\big) + c_5e^{m_5x} + \cdots + c_ne^{m_nx}$."* S002·13

Key points

1. A root repeated k times gives $(c_1 + c_2x + \cdots + c_kx^{k-1})e^{mx}$: the highest power is x^{k-1}, not x^k. ESA ★★★ S002·12 · T002 00:15:15
2. Build the CF block by block, one block per distinct root (or conjugate pair), then number the constants $c_1, c_2, \ldots$ straight through. ESA ★★★ T002 00:16:37 · T002 00:18:09
3. A repeated complex pair multiplies both $\cos\beta x$ and $\sin\beta x$ by a polynomial: $(c_1 + c_2x)\cos\beta x + (c_3 + c_4x)\sin\beta x$. ESA ★★ S002·13 · T002 00:19:14
4. Factorise higher-degree AEs by trying $m = \pm1, \pm2, \pm\frac{1}{2}, \ldots$ and synthetic division; biquadratics like $m^4 + 13m^2 + 36$ factor as quadratics in m^2. ESA ★★ (added for clarity) CW U2 P6 · CW U2 P9

Exam traps

Exam trap. For a triple root write up to x^2, not x^3. The slide's case (iii) has four equal roots, hence up to x^3. S002·12 · T002 00:15:15

Exam trap. In case (iv) the second complex pair takes new constants c_3, c_4. The lecture notes that an older version of the slide reused c_1, c_2 (T002 00:18:09); the current slide 002·13 is correct. S002·13 · T002 00:18:09

4 Solving problems, with initial conditions

Key points

1. With initial conditions, find the general solution first, then differentiate it as often as the conditions need, then substitute. ESA ★★ CW U2 P3 · CW U2 P8
2. Conditions need not be at $x = 0$: CW P9 gives $y(\frac{\pi}{2})$ and $y'(\frac{\pi}{2})$ (a boundary-value problem). ESA ★★ CW U2 P9
3. The independent variable need not be x: CW P5 is $\frac{d^4x}{dt^4} + 4x = 0$, so the CF is in t. ESA ★ CW U2 P5

Exam traps

Exam trap. Differentiate the whole general solution (product rule on $e^{2x}\cos x$) before substituting $x = 0$; $y'(0) \neq c_2$ in CW P3. ADDED FOR CLARITY CW U2 P3

★ Past-paper questions

Eight past ESA papers were checked: UE22MA141A Dec 2023, Jul 2023 and May 2024, and UE23MA141A Jan 2024 sets 1–4 and June 2024. No paper asks a homogeneous equation on its own (0 of 8), so no question is about this class alone. But every Unit 2 ESA question is non-homogeneous and needs a CF first: 24 occurrences in the 8 papers, all of them (table C4.2). The CF cases appear as follows over the distinct questions: real distinct roots 6, real repeated roots 7 (including two double pairs in Jan 2024 set 4 Q2.c and a zero root with a double root in set 3 Q2.c), complex roots 6 (of 19 distinct questions). Stars follow that count: the three basic cases ★★★; two complex pairs, a repeated complex pair and initial conditions ★★ (class-work only); a complex pair repeated three times ★ (lecture only).

2 · Second-order equations: three cases

ESA Jan 2024 · 6 marks · UE23MA141A set 3, Q3.b (CF part) · solved in W2.5

Series circuit: $R = 40\,\Omega$, $L = 1$ H, $C = 16 \times 10^{-4}$ F, $E = 100\cos 5t$. Find the charge at any time t. (CF: complex roots.)

ESA Dec 2023 · 7 marks · UE22MA141A Q3.a (CF part; also May 2024 Q3.b, June 2024 Q2.a) · solved in W2.2

Solve $(D-2)^2 y = 8(e^{2x} + \sin 2x)$. (CF: $(c_1 + c_2 x)e^{2x}$, a double root.)

3 · Equations of order n: combining the cases

ESA Jan 2024 · 6 marks · UE23MA141A set 4, Q2.c (CF part) · solved in W3.5

Solve $[(D-1)^2(D+1)^2]y = \sin^2 \frac{x}{2} + 2^{-x} + x$. (CF: two double roots.)

ESA Jan 2024 · 6 marks · UE23MA141A set 3, Q2.c (CF part) · solved in W3.6

Solve $(D^3 - 2D^2 + D)y = x^2 e^{-x} + \sin^2 x$. (CF: a zero root and a double root.)

4 · Solving problems, with initial conditions

ESA Jan 2024 · 5 marks · UE23MA141A set 2, Q2.d (CF part) · solved in W4.3

Solve $(D^3 - 6D^2 + 11D - 6)y = e^{2x} + x$. (CF: roots 1, 2, 3, as in CW P8.)